

Meetings

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Terriermon Assistant has n meetings he needs to attend. Each meeting has a start time (s_i) and end time (e_i).

If one meeting ends at the start time of another, he can instantly change to the next meeting. However, it is impossible to attend two meetings that overlap for any period of time.

Determine whether it is possible for him to attend all the meetings.

Input

The first line of input consists of a single integer n ($1 \leq n \leq 100,000$), the number of meetings.

n lines follow, the i th containing two integers s_i and e_i ($0 \leq s_i < e_i \leq 1,000,000$), representing the start and end time of the i th meeting.

Output

If Terriermon Assistant is able to attend all meetings, display **YES**. Otherwise, display **NO**. Case will be ignored.

Examples

standard input	standard output
3 0 1 2 3 4 5	Yes
4 3 7 8 11 1 2 12 20	Yes
3 1 10 5 15 20 25	No
3 0 1 1 2 2 3	Yes

Note

In sample case 1, Terriermon Assistant can attend all the meetings as no meetings collide with each other. He attends the meetings in the order they are in. Meeting 1 starts at time 0 and ends at time 1, then meeting 2 starts at time 2 and ends at time 3, and meeting 3 starts at time 4 and ends at time 5.

In sample case 2, Terriermon Assistant can attend all the meetings. He attends the meetings in order 3, 1, 2, 4. Meeting 3 starts at time 1 and ends at time 2, then meeting 1 starts at time 3 and ends at time 7, then meeting 2 starts at time 8 and ends at time 11, and meeting 4 starts at time 12 and ends at time 20.

In sample case 3, Terriermon Assistant has 2 meetings that will collide as they overlap over the time period from 5 to 10. These are meetings 1 and 2 as meeting 1 starts at time 1 and ends at time 10 while meetings 2 starts at time 5 and ends at time 15.

In sample case 4, no meetings will collide as although two meetings have start and end times that are equal, he can immediately change to the next meeting.